

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	The energy of the photon becomes (1) PE and KE of electron (accept freeing electron for PE) (1) Alternative: Photon energy = KE + workfunction (1) Photon energy = KE + energy to release electron (2)	2			2		
		(ii)	Current / no. of electrons per sec / charge per sec etc. Accept power			1	1		
	(b)		Substituting correctly for r in equation 3 e.g. $p^2 = \frac{m_e e^2}{4\pi\epsilon_0 \frac{nh}{2\pi p}} (1)$ Correct algebra leading to $p = \frac{m_e e^2}{2\epsilon_0 nh} (1)$ (Minimum algebra: $p^2 = \frac{m_e e^2}{4\pi\epsilon_0 \frac{nh}{2\pi p}}$ QED i.e. cancelling π and p)		2		2	2	
	(c)	(i)	Minimum of 3 values substituted correctly (1) Answer = 13.55 [eV] or final substitution seen (1)	1	1		2	2	
		(ii)	Explain $n = 1$ e.g. leaving $n = 1$ or substituting $n = 1$ (1) Explain $n = \infty$ or PE at ∞ (1)		2		2		
	(d)		Valid method adopted e.g. using $n = 3$ and $n = 2$ to calculate the wavelength OR using 656 nm to confirm energy gap between $n = 2$ and 3 (1) $E = \frac{hc}{\lambda} = 3.03 \times 10^{-19}$ [J] OR $13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right)$ OR 1.89 (1) Final confirmation seen e.g. $3.03 \times 10^{-19} \text{ J} = 13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}$ OR $\frac{hc}{13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}} = 656 \text{ nm} (1)$			3	3	2	

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	(e)		Easier to escape or more electrons escaping or tunnelling linked to electrons escaping (1) Light at lower V or <u>measured</u> V is lower or actual barrier is higher (1)		2		2		
	(f)	(i)	Electrons have greater [internal] energy i.e. link between temperature and energy (1) More electrons (1)		2		2		
		(ii)	Use of $\frac{3}{2}kT$ can be implied (1) Conversion to eV (1) Answer = 0.038 [eV] (1) Diode switches on 0.038 V too early or 0.038 V is an uncertainty - any good sensible comment regarding this equivalent pd (1)			4	4	2	
			Question 8 total	3	9	8	20	8	0